MATH 110 — Week 3

Homework 3 — Solutions

HW3 P1(a).

Idea. The preimage inherits the subspace test because T preserves combinations.

1. $T0 = 0 \in U$, so zero lies in $T^{-1}(U)$.
2. If $Tv, Tw \in U$, then $T(av + w) = aTv + Tw \in U$. Thus $T^{-1}(U)$ is a subspace.

HW3 P1(b).

Idea. Restrict the domain and codomain so the desired intersection is precisely the range.

1. Define $R : T^{-1}(U) \to U \cap \operatorname{range} T$ by $Rv = Tv$. The output is in U by the domain condition and in $\operatorname{range} T$ by its form. The map is linear by restriction.
2. Every $w \in U \cap \operatorname{range} T$ is Tv for some v, and $Tv \in U$ implies $v \in T^{-1}(U)$; thus R is onto.
3. Since $0 \in U$, every element of $\ker T$ belongs to the restricted domain, giving $\ker R = \ker T$. Apply rank–nullity to R on the finite-dimensional subspace $T^{-1}(U)$: $\dim T^{-1}(U) = \dim \ker R + \dim \operatorname{range} R = \dim \ker T + \dim(U \cap \operatorname{range} T)$.

HW3 P1(c).

Idea. Use $U = \operatorname{span}(1, x^2)$ to express the derivative condition as a preimage.

1. Example 25 gives $\ker D = \operatorname{span}(1)$ and $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$. Thus $U \cap \operatorname{range} D = U$ has dimension 2.
2. Part (b) gives $\dim D^{-1}(U) = 1 + 2 = 3$. Direct integration gives $p = c + bx + (a/3)x^3$, so a confirming basis is $1, x, x^3$.

HW3 P2.

Idea. A common kernel and range must have half the domain dimension.

1. On $\mathbb{R}^4$ let $T(x_1, x_2, x_3, x_4) = (x_3, x_4, 0, 0)$. This is linear coordinatewise; both its kernel and range are $\operatorname{span}(e_1, e_2)$.
2. If such a map existed on $\mathbb{R}^5$, rank–nullity would give $5 = 2 \dim \ker T$, impossible because dimension is an integer.

HW3 P3.

Idea. The constraints define the two outputs, and simple polynomials give both coordinate directions.

1. Let $S(p) = (p(1), p'(1))$. Evaluation and differentiation are linear, and its kernel is exactly U.
2. $S(1) = (1, 0)$ and $S(x - 1) = (0, 1)$, so S is onto and has rank 2. Rank–nullity gives $\dim U = 5 - 2 = 3$.
3. This agrees with HW2: the basis $(x - 1)^2, (x - 1)^3, (x - 1)^4$ has length 3.

HW3 P4.

Idea. Linearity preserves any fixed relation, including its nonzero coefficients.

1. If $\sum a_j v_j = 0$ with not all $a_j = 0$, applying T gives $\sum a_j T v_j = 0$ with those same coefficients. Thus the image list is dependent.
2. For the zero map $T : \mathbb{R} \to \mathbb{R}$, the independent list (1) maps to (0), which is dependent.

HW3 P5(a).

Idea. Each column consists of the monomial coefficients of one basis image.

1. $T1 = 1$, $Tx = x + x^2$, $Tx^2 = x^2 + 2x^3$, $Tx^3 = x^3 + 3x^4$.
2. With the prescribed bases, $A = M(T) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

HW3 P5(b,c).

Idea. Compare output coefficients to check independence of the image columns.

1. For $p = a + bx + cx^2 + dx^3$, $Tp = a + bx + (b + c)x^2 + (2c + d)x^3 + 3dx^4$. If this is zero, the constant and linear coefficients give $a = b = 0$, then the quadratic gives $c = 0$, and the cubic gives $d = 0$. Thus $\ker T = \{0\}$, nullity 0.
2. The four images above span the range by linearity, and their independence follows from the same coefficient argument. Hence the range has basis $1, x + x^2, x^2 + 2x^3, x^3 + 3x^4$ and dimension 4.
3. The monomial coordinate vector of Tp is $A(a, b, c, d)^\mathsf{T}$, an arbitrary combination of its four columns. Thus the coordinate range is exactly their span; the polynomials themselves are recovered from their coordinates.

HW3 P6.

Idea. In $\mathbb{F}_2$, each element is its own additive inverse.

1. The second equation gives $z = x$. The first becomes $y + w = 1$, so $w = 1 + y$.
2. Choose $x, y \in \{0, 1\}$ freely. The four solutions (x, y, z, w) are $(0, 0, 0, 1)$, $(0, 1, 0, 0)$, $(1, 0, 1, 1)$, $(1, 1, 1, 0)$. Substitution verifies both equations modulo 2.

HW3 P7 (optional).

Idea. Translation by a known solution gives a bijection with the kernel.

1. Choose x_0 with $Ax_0 = b$. Then $Ax = b$ iff $A(x - x_0) = 0$, so all solutions are uniquely $x_0 + u$ with $u \in \ker A$.
2. By rank–nullity, $\dim \ker A = m - \operatorname{rank} A$. In a kernel basis every coefficient is independently 0 or 1; uniqueness of basis coordinates gives exactly $2^{m - \operatorname{rank} A}$ solutions.

Exam practice, Question 2(a).

Idea. Injectivity converts a relation among images into a relation among the basis vectors.

1. If $\sum a_j Tv_j = 0$, then $T(\sum a_j v_j) = 0$. Injectivity gives $\sum a_j v_j = 0$ by the zero-kernel criterion.
2. Independence of $v_1, \ldots, v_n$ forces every $a_j = 0$.

Exam practice, Question 2(b).

Idea. Extend the independent images to a basis so Proposition 21 can define S freely.

1. Extend $Tv_1, \ldots, Tv_n$ to a basis $Tv_1, \ldots, Tv_n, w_1, \ldots, w_r$ of W using Week 2 Proposition 11 (extension of an independent list).
2. Prescribe $S(Tv_j) = v_j$ and $S(w_k) = 0$. Proposition 21 gives a unique linear S on W.
3. For $v = \sum a_j v_j$, $STv = \sum a_j S(Tv_j) = \sum a_j v_j = v$. Hence $ST = I_V$.